

Chapter 65: Dermatological Disorders — Revision Notes

Williams Obstetrics 26e, pp. 3008–3024 of the PDF. High-yield numbers are in **bold**. The three tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 4 figures are included from the PDF.

Big picture: Four dermatoses are unique to pregnancy and all itch. Only **pemphigoid gestationis** and **intrahepatic cholestasis** harm the fetus. Most other skin disease is treated with emollients, antihistamines and low/moderate-potency topical steroids.

1. Pregnancy-Specific Dermatoses

- The four: **pemphigoid gestationis (PG)**, **PUPPP**, **atopic eruption of pregnancy (AEP)** and **intrahepatic cholestasis of pregnancy**.
- They look alike and **pruritus is common** to all.
- **Adverse fetal outcomes only with PG and cholestasis.**
- Cholestasis (Ch 58): **no primary skin lesions**; pruritus with ↑ **serum bile acids** and mildly ↑ transaminases.

Table 65-1. Pregnancy-specific Dermatoses

Disorder	Frequency	Characteristic Lesion	Adverse Pregnancy Effects	Treatment
PUPPP	Common	Erythematous pruritic papules or plaques; patchy or generalized on abdomen, thighs, buttocks, especially within striae, but with umbilical sparing	None	Oral antipruritics, emollients, topical corticosteroids, and if severe, oral steroids
AEP			None	(as above)
– Eczema in pregnancy	Common	Dry, red scaly patches on extremity flexures, neck, face		(as above)
– Prurigo of pregnancy	Common	1–5 mm pruritic red papules on extensor surfaces, trunk		(as above)
– Pruritic folliculitis of pregnancy	Rare	Small red papules, sterile pustules on trunk		(as above)
Pemphigoid gestationis	Rare	Erythematous pruritic papules, plaques, vesicles, and bullae; abdomen often with umbilical involvement, extremities	Preterm birth, fetal-growth restriction, transient neonatal lesions	(as above)
Cholestasis of pregnancy	Common	No primary lesions, secondary excoriations from scratching	Increased perinatal morbidity	Oral antipruritics, cholestyramine, ursodeoxycholic acid

AEP = atopic eruptions of pregnancy; PUPPP = pruritic urticarial papules and plaques of pregnancy. The table gives prurigo papules as 1–5 mm; the chapter text says 5–10 mm.

Pemphigoid gestationis (PG)

- **Rare autoimmune bullous disease.** Formerly "herpes gestationis", but **not related to herpesvirus**.

- **Lesions:** pruritic papules and urticarial plaques → **vesicles/bullae after 1–2 wk.**
 - **Involves the umbilicus; spares mucous membranes, scalp and face.**
- **Pathogenesis:** maternal **IgG against collagen XVII (= BP180)** in the basement membrane of skin and amnion → complement activation → eosinophil chemotaxis and degranulation → dermal–epidermal junction damage → blisters.
- **Course:**
 - Usually starts in a **first pregnancy**, in the **2nd or 3rd trimester**. **Postpartum onset or flare is common;** antepartum flares and remissions.
 - **Recurs** in most later pregnancies, **earlier and more severely.**
 - Associated with other autoimmune disease, especially **Graves disease.**
 - Early onset + blistering → **preterm birth, oligohydramnios, FGR** (mild placental insufficiency from IgG/complement on the amnionic basement membrane) → surveillance is reasonable.
 - **Neonatal lesions in 5–10%** (passive IgG): wound care only, clear in a few weeks.
 - Maternal lesions heal **without scarring**; most are disease-free **by 6 months**. May flare with **menses or oral contraceptives.**
- **Differential:** PUPPP (before bullae form), bullous pemphigoid, pustular psoriasis, erythema multiforme, linear IgA dermatosis, urticaria, dermatitis herpetiformis, contact dermatitis, AEP. **Exclude SJS/TEN.**
- **Diagnosis:**
 - **Most accurate: immunofluorescence of a skin punch biopsy** → **C3** (± IgG) along the basement membrane zone.
 - Serum **anti-BP180 IgG by ELISA.**
- **Treatment:**
 - Early: **high-potency topical steroids + oral antihistamines.**
 - Refractory: **oral prednisone/prednisolone** (inactivated by placental 11β-hydroxysteroid dehydrogenase → low fetal exposure), e.g., **0.5–1 mg/kg/d**, then taper.
 - Rare intractable cases: IVIG, cyclosporine, plasmapheresis. **Rituximab** early only; later use risks transient newborn immunosuppression.

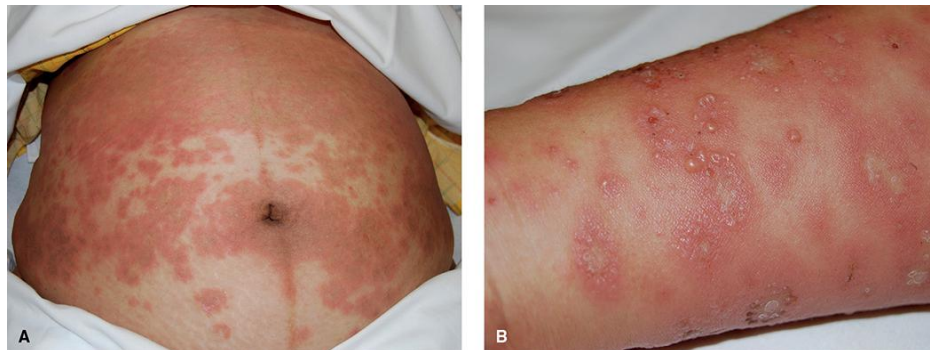


Figure 65-1. Pemphigoid gestationis: (A) abdominal plaques involving the umbilicus; (B) blisters on the wrist and forearm.

Pruritic urticarial papules and plaques of pregnancy (PUPPP)

- Also called **polymorphic eruption of pregnancy**. **Common.** Appears **late in pregnancy** (rarely postpartum).
- Intensely pruritic **1–2-mm erythematous papules** → coalesce into urticarial plaques.
- **Abdomen and proximal thighs in 97%.** Starts **within striae** and **spares the periumbilical skin**. Face, palms and soles are rarely involved.
- **Risk factors:** white race, **nulliparity**, **multifetal gestation**, **male fetus.**
- **Seldom recurs.** Cause unknown; **not autoimmune.**
- **Differential:** contact dermatitis, drug eruption, viral exanthem, insect bites, scabies, pityriasis rosea, other pregnancy dermatoses — especially **pre-bullous PG** (skin biopsy + **negative BP180 antibodies** separate them).
- **Treatment:** oral antihistamines, emollients, topical steroids; a few need short-course systemic steroids.

- **No adverse pregnancy effects.** Resolves within days of delivery without scarring; persists **2–4 wk postpartum** in **15–20%**.



Figure 65-2. PUPPP: small papules on the buttock and thigh and within abdominal striae (umbilicus spared).

Memory hook: PG goes **for** the umbilicus; PUPPP **P**asses over it.

Atopic eruption of pregnancy (AEP)

- Umbrella term for three conditions: **eczema in pregnancy**, **prurigo of pregnancy** (prurigo gestationis) and **pruritic folliculitis of pregnancy**.
- $\frac{2}{3}$ **eczematous changes**, $\frac{1}{3}$ **papular lesions**. History of prior **atopy** helps diagnosis. **No fetal risk**.

Subtype	Features
Eczema in pregnancy	Most common pregnancy-specific dermatosis. Dry, thickened, scaly red patches on flexures, nipples, neck, face. Often $\uparrow$ serum IgE .
Prurigo of pregnancy	Itchy erythematous papules/nodules (5–10 mm in text) on extensor surfaces and trunk
Pruritic folliculitis	Rare. Small erythematous follicular papules and sterile pustules , mainly on the trunk

- **75% start in the 1st or 2nd trimester.** Resolve with delivery but can persist **up to 3 months** postpartum. Recurrence variable but common.
- **Diagnosis of exclusion:** PG biopsy/serology negative; **bile acids** exclude cholestasis.
- **Treatment:**
 - Chronic **emollients**, as-needed oral antihistamines, **low/moderate-potency topical steroid twice daily** for several weeks.
 - Maintenance: moisturizer alone or topical steroid once or twice **weekly**.
 - Severe eczema: short-course ultrapotent topical steroid; sometimes topical tacrolimus, **narrow-band UVB**, oral steroid or cyclosporine.

2. Other Dermatological Conditions

Acne vulgaris

- May worsen, improve or stay stable.

- **Topical first line: benzoyl peroxide 2.5–5% or azelaic acid 15–20%.**
 - Or **5% benzoyl peroxide + topical erythromycin or clindamycin** (benzoyl peroxide limits *P. acnes* resistance).
- **Topical retinoids** (tretinoin, adapalene): no ↑ major defects in large studies, but **best avoided**, especially in the 1st trimester. **Tazarotene is contraindicated.**
- **Severe:** oral **erythromycin, azithromycin, cephalexin or amoxicillin** + benzoyl peroxide. Ideally **delay to the 2nd trimester** and limit to **4–6 wk.**
- Oral steroid short course or intralesional injection for severe cases; **oral agents after the 1st trimester** (small **facial cleft** risk).

Pearl: Oral tetracyclines (doxycycline, minocycline) and oral isotretinoin — the usual non-pregnant acne mainstays — are not used in pregnancy.

Psoriasis

- Variable course; **postpartum flares are common.**
- **Stepwise treatment:**
 1. **Emollients** → add **low/moderate-potency topical steroids.**
 2. Resistant: restrained **high-potency/ultrapotent** topical steroids (2nd–3rd trimesters).
 3. **UVB phototherapy** = second line.
 4. Third tier: **cyclosporine, oral steroids, TNF-α antagonists** (adalimumab, etanercept, infliximab).
- Psoriasis itself does **not** raise adverse outcomes. **Severe disease:** small ↑ low birthweight, GDM, preeclampsia.

Pustular psoriasis of pregnancy

- Formerly **impetigo herpetiformis. Rare**; can cause **severe systemic symptoms that mimic sepsis.**
- **Lesions:** erythematous, pruritic plaques ringed by **sterile pustules** that enlarge, coalesce and crust. Start in **intertriginous areas** → torso, extremities, oral mucosa.
- **Complications:** massive **fluid loss** → **hypovolemia**, placental insufficiency, fetal compromise; secondary infection.
- **Labs:** leukocytosis, ↑ **ESR, hypocalcemia, hypoalbuminemia.**
- **Workup:** blood and skin cultures (**sterile**) + skin biopsy (**epidermal neutrophilic pustules**).
- **First-line treatment** (choice by severity):
 - **Prednisone 15–80 mg/d**
 - **Cyclosporine 2–3 mg/kg/d**
 - Infliximab
 - Topical steroids or topical calcipotriene
- **Adjuncts:** fluids, electrolyte correction, IV antibiotics for secondary infection.
- Resolves quickly postpartum → **early delivery can be considered.** Recurs in later pregnancies and with menses or OCs.



Figure 65-3. Generalized pustular psoriasis of pregnancy: erythematous plaques ringed by sterile pustules that crust.

Erythema nodosum

- **Inflammation of subcutaneous fat.** Triggers: **pregnancy**, infections, sarcoidosis, drugs, Behçet, IBD, malignancy.
- Tender, red, warm **1–6-cm nodules** on **extensor surfaces**; they flatten and change color like a bruise (red/purple → yellow-green).
- Treat the **underlying cause**. Resolves in **1–6 wk without scarring** (may leave hyperpigmentation).

Pyogenic granuloma

- Actually a **lobular capillary hemangioma**; common in pregnancy (↑ **VEGF**).
- Sites: **gingiva**, other oral/nasal surfaces, hand, after minor irritation or trauma. **Grows fast and bleeds easily.**
- **Bleeding:** pressure + **silver nitrate** or **Monsel (ferric subsulfate) paste**.
- Often **regresses within months postpartum** → may observe antepartum.
- **Excise** if symptomatic, persistent postpartum or diagnosis unclear (scalpel + suture, electrosurgical curettage, laser, cryotherapy). Oral lesions → specialist.



Figure 65-4. Pyogenic granuloma on the gingiva: lobulated red vascular growth that bleeds easily.

Miscellaneous conditions

- **Hidradenitis suppurativa**: keratin-plugged follicles → painful nodules, abscesses, **sinus tracts** (Hurley stages I–III). Usually unchanged or better in pregnancy.
 - **Stage I/II**: topical **1% clindamycin twice daily for 12 wk** (alternatives: 0.75% metronidazole, 2% erythromycin gel).
 - **Stage II/III**: oral **clindamycin 300 mg + rifampin 300 mg twice daily for 10 wk**.
 - Second line for advanced disease: **adalimumab** (anti-TNF-α).
- **Neurofibromatosis**: neurofibromas, café-au-lait spots, axillary/inguinal freckling, **Lisch nodules**, optic gliomas.
 - Neurofibromas may **grow in size and number** in pregnancy.
 - **NF1** → ↑ **preeclampsia and preterm birth**; **NF2** → possible preeclampsia risk. Prenatal genetic diagnosis is available for both.
- **Rosacea fulminans (pyoderma faciale)**: facial pustules and draining sinuses, **no comedones**. Topical/oral antibiotics ± topical steroids; sometimes short oral steroids or drainage.

3. Dermatological Treatment

Oral antihistamines (for pruritus)

Generation	Drug	Dose
First	Diphenhydramine	25–50 mg q6h
First	Chlorpheniramine	4 mg q6h
Second (less sedating)	Loratadine	10 mg daily
Second (less sedating)	Cetirizine	5 or 10 mg daily

Topical corticosteroids

- **Start with low- or moderate-potency** agents.
- **Ultrapotent** agents (e.g., **clobetasol 0.05%**): only for refractory disease, **2–4 wk, small areas**.
- Mild/moderate strength: **no adverse pregnancy outcomes**. High/ultrapotent: **small FGR risk with large cumulative doses** (still less than oral steroids).
- **Systemic absorption ↑ with**: large surface area, broken epidermal barrier, **occlusive dressing**, long duration, other absorption-enhancing topicals.

Table 65-2. Commonly Used Topical Corticosteroids in Pregnancy

Low-Potency Agents	Moderate-Potency Agents	High-Potency Agents	Ultrapotency Agents
Hydrocortisone 0.5–1% <i>cream, ointment</i>	Triamcinolone acetonide 0.1% <i>cream, ointment</i>	Triamcinolone acetonide 0.5% <i>cream, ointment</i>	Clobetasol propionate 0.05% <i>cream, ointment</i>
Dexamethasone 0.1% <i>cream</i>	Mometasone furoate 0.1% <i>cream</i>	Mometasone furoate 0.1% <i>ointment</i>	
Desonide 0.05% <i>foam, cream, ointment</i>	Betamethasone valerate 0.1% <i>cream, ointment</i>	Betamethasone dipropionate 0.05% <i>cream, ointment</i>	

Pearl: The same drug and strength can change potency class with the vehicle: mometasone 0.1% is moderate as a cream but high as an ointment.

Table 65-3. Some Dermatological Agents and Their Suitability for Pregnancy

Group	Action	Pregnancy Recommendation
Topical Agents		
Corticosteroids	Antiinflammatory	Suitable
Benzoyl peroxide	Antiinflammatory, comedolytic, antibacterial	Suitable
Azelaic acid	Antiinflammatory, comedolytic, antibacterial	Suitable
Antibiotics: clindamycin, erythromycin	Antibacterial	Suitable
Calcineurin inhibitors: tacrolimus, pimecrolimus	Antiinflammatory	Suitable
Calcipotriene	Antiinflammatory, vitamin D analogue	Possible poor ossification in animals; no human data. Probably suitable in dermatological dosages
Coal tar	Antiinflammatory; keratinocyte regulation	Possible animal mutagen/carcinogen; no human data. If used, delay until 2nd trimester
Retinoids: tretinoin, adapalene	Antiinflammatory, comedolytic	Large studies show no harm, but experts suggest avoiding
Retinoids: tazarotene	Antiinflammatory, comedolytic	Teratogenic; contraindicated
Crisaborole	Antiinflammatory	Limited pregnancy data; avoidance currently suggested
Systemic Agents		
Oral antibiotics: azithromycin, amoxicillin, cephalexin, erythromycin	Antibacterial	Suitable
Oral corticosteroids	Antiinflammatory	1st-trimester use linked to small risk of oral clefts
Cyclosporine	Immunomodulator	Low birthweight, preterm birth; nonteratogenic
TNF- α inhibitors: infliximab, adalimumab, etanercept	TNF- α inhibition	Suitable

From Bae, 2012; Briggs, 2017; Murase, 2014; Park-Wyllie, 2000; Rademaker, 2018; Robinson, 2012; Sekhon, 2018; Vestergaard, 2019.

- **Avoid in pregnancy (teratogens, Ch 8): methotrexate, PUVA (psoralen + UVA), mycophenolate mofetil, podophyllin, systemic retinoids.**
- Secondary bacterial skin infection → prompt oral antibiotics with **gram-positive coverage**.

Quick-Recall: Pregnancy-Specific Dermatoses at a Glance

Feature	Pemphigoid gestationis	PUPPP	AEP	Cholestasis
Frequency	Rare	Common	Common (eczema = most common)	Common
Timing	2nd–3rd tri; postpartum flare	Late 3rd tri	75% in 1st–2nd tri	Late (Ch 58)
Parity	1st pregnancy	Nulliparas , multifetal, male fetus	Prior atopy	—
Umbilicus	Involved	Spared	—	No primary lesion
Blisters	Yes (bullae)	No	Sterile pustules (folliculitis)	Excoriations only
Test	DIF: C3 at BMZ; anti-BP180 ELISA	Negative BP180	IgE ↑; exclusion	↑ Bile acids
Fetal risk	Preterm, FGR, neonatal lesions (5–10%)	None	None	↑ Perinatal morbidity
Recurrence	Yes, earlier and worse	Seldom	Common	—

Key Numbers to Memorize

- PG: bullae **1–2 wk** after plaques; neonatal lesions **5–10%**; disease-free by **6 months**; prednisone **0.5–1 mg/kg/d**
- PUPPP: papules **1–2 mm**; abdomen/thighs in **97%**; persists **2–4 wk** postpartum in **15–20%**
- AEP: **⅔ eczematous / ⅓ papular**; **75%** begin in 1st–2nd trimester; may last **3 months** postpartum
- Acne: benzoyl peroxide **2.5–5%**, azelaic acid **15–20%**; oral antibiotics for **4–6 wk**, from the 2nd trimester
- Pustular psoriasis: prednisone **15–80 mg/d**; cyclosporine **2–3 mg/kg/d**
- Erythema nodosum: nodules **1–6 cm**, resolve in **1–6 wk**
- Hidradenitis: clindamycin 1% topical **12 wk**; clindamycin + rifampin **300 mg each twice daily × 10 wk**
- Antihistamines: diphenhydramine **25–50 mg q6h**; chlorpheniramine **4 mg q6h**; loratadine **10 mg**; cetirizine **5–10 mg**
- Ultrapotent topical steroid: **2–4 wk**, small areas only (clobetasol **0.05%**)